

Clock Arithmetic and the Mysteries of Prime Numbers

Foreword

This article is written for my patient friends to whom I give math riddles that require knowledge of modular arithmetic, or as I often call it *clock arithmetic*. This article serves as an easy reference for the necessary concepts and requires no mathematical background. For this entire article, we will only care about which hour the hour-hand is on; no AM or PM, and no adding half an hour (or anything that isn't an increment of 60 minutes).

Note that any exercise can be skipped; they are just there for fun!

1 Clock Arithmetic: Addition and Multiplication

We all have (at least) two ideas of addition in our heads. The first one is the usual notion of adding quantities¹: If I have 4 bottles and add 10 bottles, I have 14 bottles. In other words

$$4 + 10 = 14$$

The other notion of addition we often use comes from when we are calculating what time it is. Consider a 12-hour clock. If it is 4 o'clock, in 5 hours it will be 9 o'clock. If instead we add 10 hours, it will be 2 o'clock. In other words, we just comfortably did

$$4 + 10 = 2$$

in our heads. This is certainly different from before, but it still makes sense! Using the same symbol “+” may cause confusion, so to distinguish the two types of addition, we shall write $+_{12}$ for “12 hour clock” addition, and keep $+$ as before. Thus, we can write:

$$4 + 10 = 14 \quad \text{and} \quad 4 +_{12} 10 = 2$$

Let us call the addition using $+$ *natural arithmetic* and the addition using $+_{12}$ *clock arithmetic*. As good mathematicians, we should see what similarities and differences these different ideas of additions have.

The first thing we should try to do is add a bunch of hours to each other and see what we get. Let us start by just picking two random numbers and playing around with them. If it is 2 o'clock, then if I add 5 it will be 7 o'clock. But if I add 17, it will *still* be 7 o'clock. More generally, adding any multiple of 12 will not change the result of my addition:

$$2 +_{12} 5 = 7, \quad 2 +_{12} 17 = 7, \quad 2 +_{12} 29 = 7, \quad 2 +_{12} 41 = 7 \dots$$

So adding 5, $(5 + 1 \cdot 12)$, $(5 + 2 \cdot 12)$, ... to 2 gives the same result. Similarly, if I have any number a between 1 and 12, if I add b , or I add $(b + k \cdot 12)$ for some $k = 1, 2, \dots$ I get the same result.

We would like to remember that adding a number n or adding a number $n + k \cdot 12$ gives the same result. Mathematicians usually use the symbol $\equiv_{12}$ between two such numbers:

$$5 \equiv_{12} 17 \equiv_{12} 29 \equiv_{12} \dots$$

¹A mathematician would say *discrete quantities* to distinguish between *continuous* quantities, which would describe something like water where the amount we add is on a “continuous” spectrum of possible amounts rather than “discrete” indivisible chunks of amounts

where we usually say that 5 is *equivalent* to 17, which is equivalent to 29, and so forth. Said differently, two numbers a, b are equivalent if they give you the same time on the clock-face after adding:

$$n +_{12} a = m \quad \text{and} \quad n +_{12} b = m$$

Therefore, in clock arithmetic we only need to work with the following numbers:

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$$

All these numbers are *not* equivalent to each other (ex. $3 \not\equiv_{12} 5$). Adding any two of these numbers using $+_{12}$ will give another number in the above list:

$$1 +_{12} 1 = 2 \quad 11 +_{12} 7 = 6 \quad 9 +_{12} 12 = 9$$

The last of these three examples is somehow different than the others; it is the only equation where one of the numbers on the left hand side of the addition $+_{12}$ also appears on the right hand side of the equality; namely 9 appears on both sides. In fact, if we try adding any other number to 9, we see that only adding 12 returns 9:

$9 +_{12} 1 = 10$	$9 +_{12} 5 = 2$	$9 +_{12} 9 = 6$
$9 +_{12} 2 = 11$	$9 +_{12} 6 = 3$	$9 +_{12} 10 = 7$
$9 +_{12} 3 = 12$	$9 +_{12} 7 = 4$	$9 +_{12} 11 = 8$
$9 +_{12} 4 = 1$	$9 +_{12} 8 = 5$	$9 +_{12} 12 = 9$

The natural next question to ask is if this works for any number, not just 9. Try doing this for any n between 1 and 12 (including 12), and you'll see that adding 12 to any number gives back the same number, and 12 is the *only* number that does this!

$$n +_{12} 12 = n$$

Notice that in natural arithmetic, this is exactly the behaviour of 0! Namely, for any natural number n :

$$n + 0 = n$$

With this parallel in mind, when we are doing clock arithmetic, we should think of 12 as being the same as 0. Mathematicians would even go so far as re-labeling 12 as 0 so that the 12 numbers of clock arithmetic are:

$$0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11$$

Re-labeling 12 as 0 has another advantage when we consider *multiplication* in clock arithmetic. If we are to think of 12 as being 0 in clock arithmetic, we should at least check that this behavior is also respected by multiplication. Remember that in natural arithmetic, multiplying any number n^2 by 0 gives 0:

$$n \cdot 0 = 0$$

Let us first test if this is the case by doing an example. If it is 3 o'clock and we multiply by 12, then we are adding 3 o'clock to itself 12 times, which would bring us 36 hours forward, which on a clock-face is 12 o'clock. But we already saw that we should think of 12 as being 0! Hence, we see that adding 3 to itself 12 times will give us back 0. In fact, if we start with any number n between 0 and 11, adding the number to itself 12 times will always bring us back to the 12 o'clock position! Therefore, 12 really *does* behave like 0 in clock arithmetic! If $\cdot_{12}$ represents multiplication in clock arithmetic, we just showed that multiplying by 12 is like multiplying by 0. For example:

$$3 \cdot_{12} 12 = 12$$

²For those who have contested this by asking "what if we let n be infinity", there is this article addressing exactly how to think about infinity and why we shouldn't conclude $0 \cdot \infty = 0$ without putting in more care

Which in our re-labeled system (where 12 is 0) becomes:

$$3 \cdot_{12} 0 = 0$$

Exercise 1: Multiplying by other numbers

When multiplying in natural arithmetic we get that $a \cdot b = 0$ if and only if $a = 0$ or $b = 0$. In clock arithmetic, it is now possible that $a \cdot_{12} b = 0$ but $a, b \neq 0$! Can you think of two non-zero numbers a, b where $a \cdot_{12} b = 0$?

Exercise 2: Combining Addition and Multiplication

Show that for any natural numbers a, b, c

$$a \cdot_{12} (b +_{12} c) = (a \cdot_{12} b) +_{12} (a \cdot_{12} c)$$

Mathematicians call this *distributivity*

Exercise 3: Subtraction

Subtraction works just the same as addition. Convince yourself that:

$$1 -_{12} 2 = 11$$

that is, if it is 1 o'clock, two hours ago it was 11 o'clock. Convince yourself that:

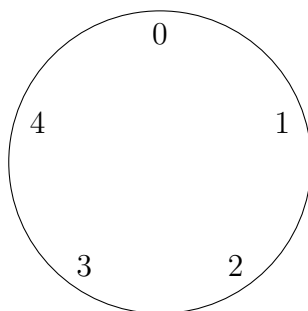
$$a -_{12} 12 = a$$

2 From Clock Arithmetic to Modular Arithmetic

So we just showed how we can think of clock arithmetic which has 12 numbers. A mathematician would then naturally ask the following question:

What if we replace 12 with any other natural number³?

Let us do exactly this and see what we get! First off, instead of writing $+_{12}$, we shall now write $+_n$. Let us call the arithmetic for all $+_n$ *modular arithmetic*. With this vocabulary, clock arithmetic is a special case of modular arithmetic. To distinguish the different modular arithmetics, we would usually say we are working *modulo* n . Clock arithmetic would be modular arithmetic modulo 12. When working modulo n , we can imagine a clock face with n positions. For example, if $n = 5$, then we can think of having a clock with 5 hours:



On this clock, $3 +_5 3 = 1$ and $2 \cdot_5 4 = 3$. When working modulo n , we will always have n numbers:

$$0, 1, 2, \dots, n-1$$

Just like for clock arithmetic, we have that we can treat n like 0.

Exercise 4: Zero in Modular Arithmetic

Let a, n be any natural numbers. For a moment, let us not label n as 0. Convince yourself that:

$$a \cdot_n n = n$$

This shows that in modular arithmetic mod n , we can *always* treat n as 0, and hence why we will always write 0 instead of n .

Exercise 5: Special Property when Modulo Prime

Remember in exercise 1 that in modulo 12, we can find two natural numbers a, b where $a \cdot_{12} b = 0$, but $a, b \neq 0$.

Convince yourself that if $n = p$ is a prime number (ex. $p = 2, 3, 5, 7, 11, \dots$) then

$$a \cdot_p b = 0 \quad \text{if and only if} \quad a = 0 \text{ or } b = 0$$

This shows that primes are somehow “special”. Mathematicians even have a different word for when $n = p$ is a prime: modular arithmetic modulo p where p is a prime is called a *finite field*.

³A natural number is of the form $1, 2, 3, \dots, n, \dots$. We in fact *can* replace it with non-natural numbers, but it took some time in history to understand what this means; this would be introduced in a course in *abstract algebra* (see[3] if you want my exposition on abstract algebra)

3 Application: Equations

So now that we defined modular arithmetic, what can we do with it? The biggest reason mathematicians care about modular arithmetic is that it allows us to give a lot more information about equations we are trying to solve!

For example, consider $x^2 + x + 1$. If I ask you to tell me whether there is an integer n , numbers of the form $\pm 1, \pm 2, \pm 3, \dots$, that solves this equation (i.e. an n such that $n^2 + n + 1 = 0$), it may take you a little while to figure out if such a number exists. A natural starting point is to just convince yourself that no small integer can solve it. By plugging in $\pm 1, \pm 2, \pm 3$ we get:

x	Calculation	Result
1	$1^2 + 1 + 1$	3
-1	$(-1)^2 + (-1) + 1$	1
2	$2^2 + 2 + 1$	7
-2	$(-2)^2 + (-2) + 1$	3
3	$3^2 + 3 + 1$	13
-3	$(-3)^2 + (-3) + 1$	7

Doing this for the first 100 integers and their negatives would give you that the equation never equals 0, however there is no guarantee that you will *never* find a big enough number n such that $n^2 + n + 1 = 0$. Using modular arithmetic, it is in fact straightforward to show that *no* solutions exist.

To show this, let's say that there *does* exist an n such that $n^2 + n + 1 = 0^4$. What we will do is show that this equation will *fail* when we consider modular arithmetic modulo 2. In modulo 2, the number n would be equivalent to:

$$n \equiv_2 0 \quad \text{or} \quad n \equiv_2 1$$

For example, if $n = -177$ (which the reader could check doesn't solve the equation, but it is just for demonstration), then:

$$-177 = 1 + 2 \cdot (-89) \equiv_2 1$$

(look at exercise 3 if the subtraction doesn't sit well with you). We can keep 1 and 0 the same because they happen to be small enough so that we don't need to shrink them. Thus, armed with the possible values of n modulo 2, we get one of two possible equations:

$$(1 \cdot 2 1) +_2 1 +_2 1 = 0 \quad \text{or} \quad (0 \cdot 2 0) +_2 0 +_2 1 = 0$$

But hang on, if we compute the left hand side of the first equation we get

$$1 +_2 1 +_2 1 = 0 +_2 1 = 1$$

which is *not* equal to the right hand side (i.e. $1 \neq 0$ and $1 \not\equiv_2 0$). The second equation is even more obvious since $(0 \cdot 2 0) +_2 0 +_2 1 = 1$ which is again not equal to 0.

So we arrived at a contradiction, because $0 \neq 1^5$. What this tells us is that we made a wrong assumption earlier in the proof, and indeed this started by assuming that there exists an integer n such that $n^2 + n + 1 = 0$. Thus, it is *impossible* that such an integer exists! Notice how easy this was to prove: we only had to check 2 possible numbers as compared to the infinite possibilities when trying to solve the equation with natural numbers!

This "reduction to finitely many cases" is one of the **key** properties of modular arithmetic that mathematicians abuse to no end.

⁴This is a usual trick called *proof by contradiction*

⁵The suspicious reader may ask why is $0 \neq 1$. In fact, there is nothing wrong with the arithmetic system where $0 = 1$, however if we let $0 = 1$ we get that every number equals 0 (convince yourself), and then nothing interesting happens.

4 Application: Patterns in Prime Numbers

Consider the equation $x^2 = p$ for some prime number p . Then obviously no natural number solves this equation, because no natural number can be the square-root of a prime⁶. However, this question gets more interesting if we instead consider it in *modular arithmetic* modulo n .

Let us demonstrate this with an example. Consider $p = 3$. Let us see if in modulo n , we can find a number m such that $m \cdot_n m = p$ (i.e. if p has a square-root modulo n). The following is a table where n goes over the first few prime numbers:

n	1 ²	2 ²	3 ²	4 ²	5 ²	6 ²	7 ²	8 ²	9 ²	10 ²
2	1									
3	1	1								
4	1	0	1							
5	1	4	4	1						
6	1	4	*3*	4	1					
7	1	4	2	2	4	1				
8	1	4	1	0	1	4	1			
9	1	4	0	7	7	0	4	1		
10	1	4	9	6	5	6	9	4	1	
11	1	4	9	5	*3*	*3*	5	9	4	1

Let us focus on the cases where $n = p$ is a prime number. Notice that $5 \cdot_{11} 5 = 3$, since $5 \cdot 5 = 25 = 3 + 2 \cdot 11$. Hence, for a mysterious reason, there *does* exist a square-root of 3 in modulo 11, and there *doesn't* appear to be a square-root of 3 modulo the other small primes like 5 and 7. There in fact exists many other prime numbers n such that 3 has a square-root modulo n , infinitely many of them in fact. But why 11? Why was the prime 3 somehow linked to the prime 11?

This is a rather fascinating result: primes somehow have *relations* between each other! Two primes are somehow tied to each other depending on whether we can take the square root of one modulo the other! Euler in the 19th century had the exact same realization, and did thousands of such computations looking for patterns. It turns out to be a rather hard problem, and it wasn't until Gauss around 50 years later that gave a definitive and satisfactory explanation how such primes relate via square-rooting: the result is now called *quadratic reciprocity*. Unfortunately, answering this question would require a lot more than what this article covers (all the details are in [3], and requires a good deal of build-up before being proven in chapter 22.4). However, I strongly encourage the reader play around with this; after a few hours or a few days, the reader will most likely stumble upon the same pattern that many other mathematicians who worked on the problem (including Euler) have discovered!

⁶This is because the square-root of a prime number is *irrational*; see [2] for a friendly introduction to this notion

5 Final Mystery

The problem in the above section turned out to be one of the first cornerstones in number theory, as it somehow connects different prime numbers to each other via equations; that is, somehow prime numbers are not completely “random” but can have subtle connections that are not obvious at first glance!

This next and final problem we shall present:

Are there infinitely or finitely many primes p such that

$$p - 3 \equiv_5 0$$

where 3 and 5 can be replaced with any numbers a, b where $\gcd(a, b) = 1$ (i.e. they have no common factors).

This question was pondered by Dirichlet in 1837, and essentially is asking about how prime numbers are “distributed” (i.e., is there a meaningful way in which we can group prime numbers together). Just like the above problem, I strongly encourage the reader to think about this problem for a couple of hours and just see what they can come up with. The result has been proven, and a full proof requires a good deal more than what can be covered in an expository article like this (though for completeness, a proof can be found in [1]).

References

- [1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Class Field Theory*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_class_field_theory.pdf.
- [2] Shay Fuchs. *Introduction to Proofs and Proof Strategies*. Cambridge University Press. URL: <https://www.cambridge.org/highereducation/books/introduction-to-proofs-and-proof-strategies/3E17FC9C3CA060246BF1B5476687AEB0#overview>.
- [3] Nathanael Chwojko-Srawley. *Everything You Need To Know About Undergraduate Algebra*. 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebra.pdf.